

Locomotion & Movement

SHORT NOTES

Locomotion and Movement

- ❖ **Locomotion:** Voluntary movements resulting in change in place/location.
- ❖ **Movement:** Significant feature of living beings.
- ❖ All locomotions are movements but all movements are not locomotion.

Table: Types of Movement/Locomotion

Type	Structure	Examples and Functions
Amoeboid	Protoplasmic streaming	• They engulf, digest food and foreign particles. • Leucocytes, macrophages and <i>Amoeba</i>.
Ciliary	Cilia	• Removing dust particles from trachea. • Passage of ova through female reproductive tract.
Flagellar	Flagella	• Water current in canal system of sponges. • Locomotion in <i>Euglena</i>. • Swimming of spermatozoa.
Muscular	Muscles	• Movement of limbs, jaws, tongue, etc.

Muscles

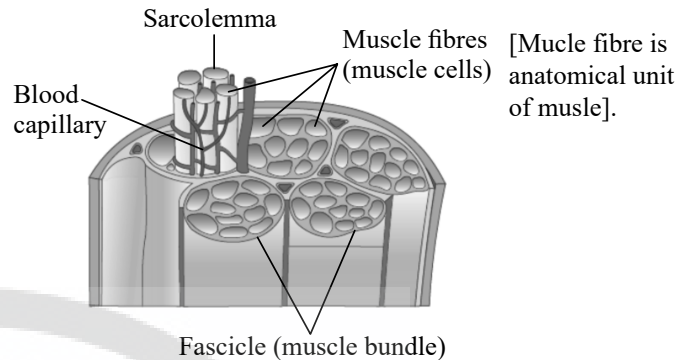
- ❖ They are mesodermal in origin.
- ❖ 40-50% of the body weight of a human adult is contributed by muscles.
- ❖ **Properties:** Excitability, contractility, extensibility and elasticity.

Table: Types of muscles based on appearance, nature of regulation and their location

Muscles	Appearance	Regulation	Location
1. Skeletal	Striated	Voluntary	Muscles of limbs.
2. Smooth	Non-striated/smooth	Involuntary	Inner walls of visceral organs.
3. Cardiac	Striated	Involuntary	Muscles of heart.

Skeletal Muscle

- ❖ Closely associated with the skeletal components of the body.
- ❖ Primarily involved in locomotion and body posture.



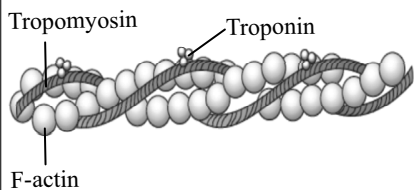
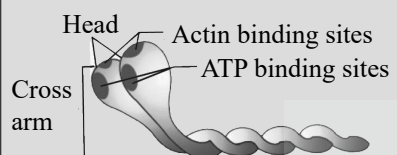
Skeletal muscle
 Many ↓
 Muscle bundles/Fascicles
 Held by ↓
Fascia
 (collagenous connective tissue)

Table: Types of muscle fibres

Features	Red	White
Myoglobin	↑	↓
Mitochondria	↑	↓
SR	↓	↑
Respiration	Mainly aerobic.	Mainly anaerobic.

- ❖ Each muscle bundle contains a number of muscle fibres.
- ❖ Muscle fibre is a syncytium (sarcoplasm contains many nuclei).
- ❖ Endoplasmic reticulum, i.e., sarcoplasmic reticulum of muscle fibre is the store house of calcium ions.
- ❖ Each muscle fibre has many parallelly arranged myofibrils/myofilaments.
- ❖ Each myofibril has dark and light bands due to actin and myosin distribution that establish striated appearance.

Table: Myofilaments and structure of contractile proteins

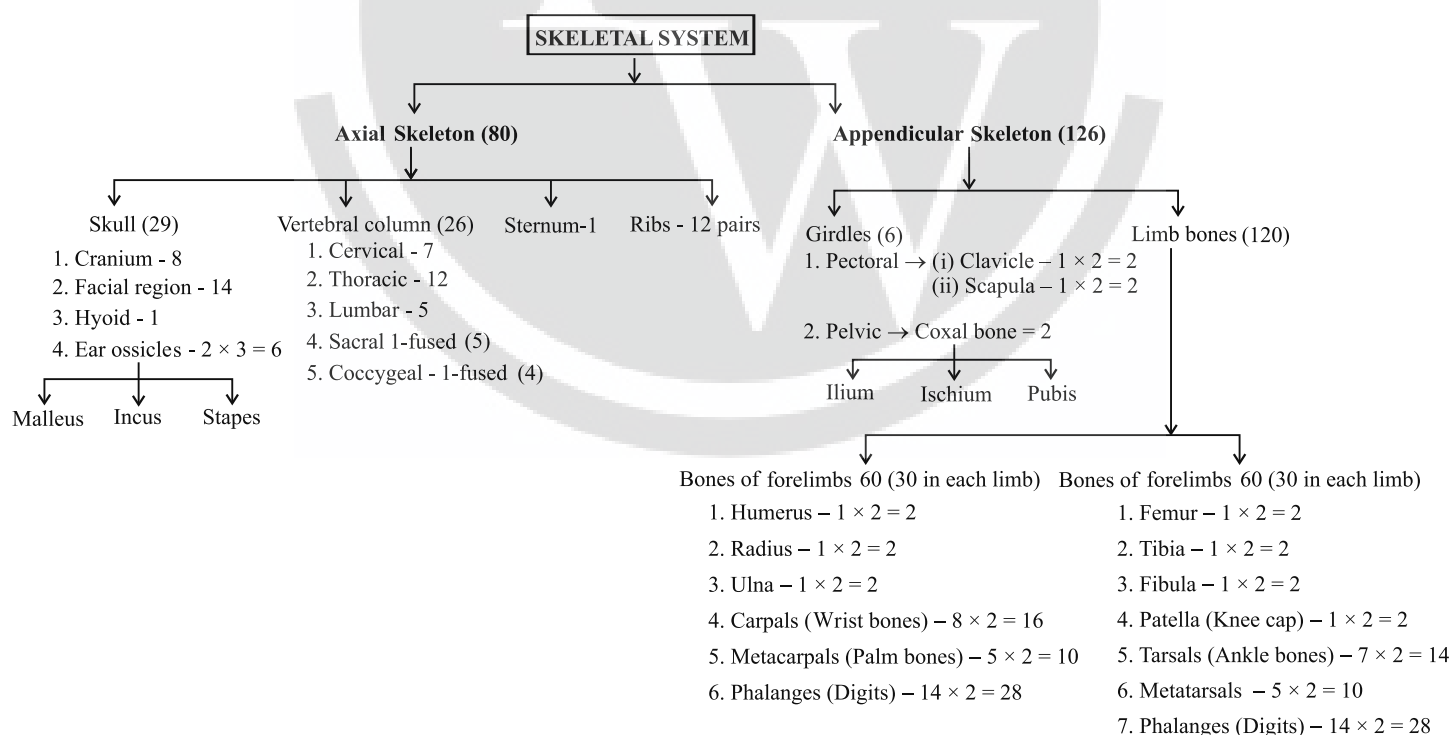
Filament	Held by	Protein
❖ Thin 	❖ Z-line (bisects I-band)	❖ Actin (contractile) ❖ Made of two 'F' actins (polymer of 'G' actin) ❖ Tropomyosin ❖ Troponin
❖ Thick 	❖ M-line (thin fibrous membrane)	❖ Myosin (contractile) ❖ Monomeric protein - meromyosin. <pre> graph TD Meromyosin --> LMM["LMM (Light)"] Meromyosin --> HMM["HMM (Heavy)"] LMM --> Tail HMM --> Head HMM --> Short_arm["Short arm"] </pre>

Mechanism of Muscle Contraction

- ❖ Contraction of muscle fibre takes place by the sliding of the thin filaments over the thick filaments.
- ❖ A motor neuron + Muscle fibres = Motor unit

Skeletal System

- ❖ Framework of 206 bones & few cartilages.
- ❖ Principle division:
 - + Appendicular skeleton.
 - + Axial skeleton.



- ❖ **Scapula**, expanded to form acromion process that articulates with clavicle.
- ❖ **Glenoid cavity** in scapula articulates with humerus head to form shoulder joint.
- ❖ **Acetabulum**, articulates with femur to form hip joint.
- ❖ 2 halves of pelvic girdle meet ventrally to form **pubic symphysis** containing **fibrous cartilage**.

Joints

- ❖ They are essential for all types of movements.
- ❖ Force generated by muscle is used to carry out movement through joints, where joint acts as **fulcrum**.

Table: Types of Joints

Types	Bones Joined by	Movement	Examples
Fibrous	Dense fibrous connective tissue.	Do not allow any movement.	Flat skull bones fused end to end via sutures to form cranium.
Cartilaginous	Cartilage.	Limited movement.	Adjacent vertebrae.
Synovial	Fluid filled synovial cavity between 2 bones.	Considerable movement, helps in locomotion and many other movements	Humerus and pectoral girdle (ball and socket joint), knee joint (hinge joint), atlas and axis (pivot joint), between carpals (gliding joint), Carpal and metacarpal of thumb (Saddle joint).

Table: Disorders of muscular and skeletal system

Disease	Causes	Impact
Myasthenia gravis	Autoimmunity	• Affect neuromuscular junction. • Fatigue, weakening and paralysis of skeletal muscles.
Muscular dystrophy	Genetic	• Progressive degeneration of skeletal muscles.
Tetany	Low Ca^{2+} in body fluid.	• Rapid spasm in muscles (wild contractions).
Arthritis	Inflammation.	• Inflammation of joints.
Gout	Accumulation of uric acid crystals.	• Inflammation of joints.
Osteoporosis	❖ Age related. ❖ Decreased levels of estrogen.	• Decreased bone mass and increased chances of fracture.